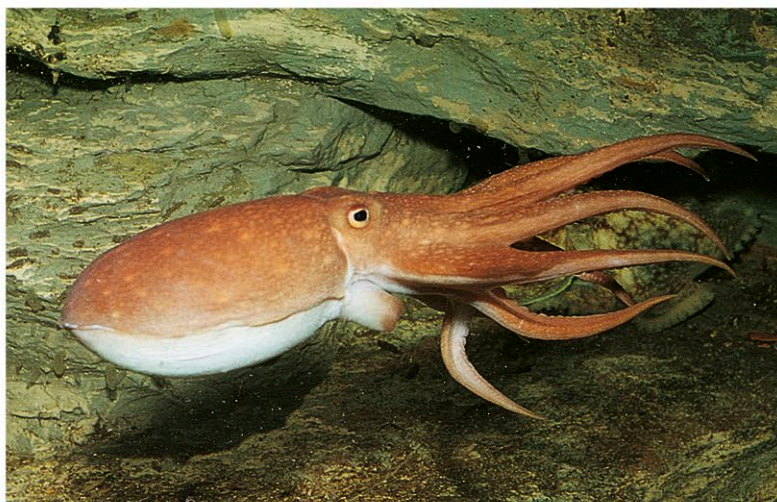




BACK TO FRONT

Octopuses are bilaterally symmetrical. They have a head with a pair of eyes—one on each side. Although the arms look as if they are radially symmetrical they are not—they are arranged around the mouth. When an octopus swims, it is the rear part of the body that leads.

worm, the adult body consists of exactly 959 cells. Each one is the product of a precise sequence of cell divisions and ends up in a location encoded in the animal's genes.

On a larger scale, one of the fundamental differences between different invertebrates lies in the symmetry of their bodies. Animals in some groups—such as sea anemones—have radial symmetry. Their body parts are arranged in the same way as the spokes or rim of a wheel, usually with the mouth at the center. These animals are often fixed in place, but if they do move, any part of the “wheel” can lead. By contrast, animals with bilateral symmetry usually have a distinct head, which leads the way when they move. These animals have an imaginary central dividing line, which splits them into 2 more or less equal parts.

During the course of evolution, some invertebrates have developed a combination of these body plans. Adult starfish, for example, show almost perfect radial symmetry, except for a small off-center opening (or madreporite) that connects their water vascular system to the outside.

Internally, invertebrates show a wide range of body plans. Sponges—uniquely in the animal kingdom—are full of perforations, or pores. By drawing water through these pores, they filter out tiny particles of food. Some simple invertebrates have a baglike anatomy, with a mouth that leads to a central digestive cavity. This arrangement, seen, for example, in cnidarians and flatworms, has one major drawback: indigestible waste has to be ejected through the mouth. Most invertebrates avoid this problem by having a digestive tract that runs through the body, in effect forming a tube. This plan—which is also found in vertebrates—allows the digestive system to work like a production line, with different parts being specialized for separate tasks.

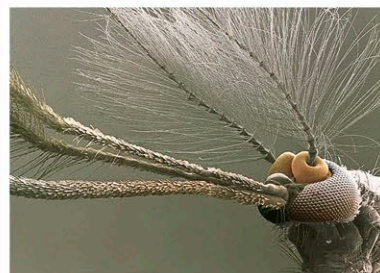
Although invertebrates do not have bones, many have a skeleton, which protects and supports them. The

skeleton may be built of hard parts, but it may also work by fluid pressure (see panel, opposite). Known as a hydrostatic skeleton, this way of supporting a soft body is found only in the invertebrate world.

Senses

For those invertebrates that are permanently fixed in one place, the sensory world is relatively simple. These animals reach out toward anything that gives off the scent of food, and retreat from anything that might be a threat. They are sensitive chiefly to dissolved chemicals, direct contact, or changes in pressure. These animals have specialized nerve cells, or receptors, scattered throughout the body, and a simple nervous system without a brain.

This system works well enough for organisms that rarely move, but mobile invertebrates need more sensory information. They have more elaborate nervous systems, with sense organs to gather information and a brain to process it and trigger the most suitable



TUNING IN

A male mosquito's bushy antennae are used for hearing. This is an example of hearing of a precise kind: the antennae respond to the exact frequency of the female's whining flight. The frequency used varies from one species to another.

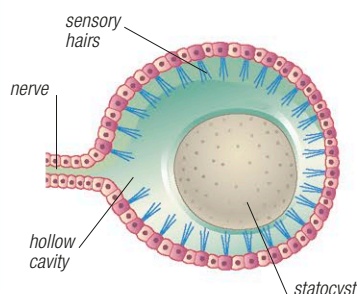
MULTIPLE ATTRACTION

Attracted by color and airborne odors, flies congregate on ripe fruit. They taste this sugary food with their mouthparts, and with chemical receptors on their feet.

response. In some of these animals, such as flatworms, the brain is tiny, but in octopuses and other cephalopods, it is highly developed and can be bigger than that of some vertebrates.

Vision is often one of the most important senses for invertebrates that move. Some of them, such as flatworms, have rudimentary eyes without lenses, which simply detect changes in the overall level of light. But many invertebrates have much more complex eyes, equipped with lenses that produce an image. One form, known as a compound eye, is widespread among arthropods, including insects. A compound eye consists of many self-contained units, each with its own receptors and lens system. Each unit responds to part of the field of view, and the total number of units determines how much detail the animal sees. A smaller number of invertebrates—principally cephalopods—have eyes that resemble our own. In these, a single lens focuses light onto a sensitive surface, the retina, which converts the image into signals that travel to the brain.

For invertebrates that live on land, hearing is also a key sense, partly because it is often used for locating a



DETECTING GRAVITY

Bivalves have microscopic sensors that respond to the direction of gravitational pull. The sensor consists of a chamber lined with nerve cells that detect the movement of a mineral weight (known as a statocyst).



MIMICRY

Mimicry occurs in the vertebrate world, but invertebrates are the true experts at this form of deception. Invertebrates mimic various objects, from pebbles and twigs to bird droppings, largely to avoid being eaten. They also mimic each other, either as a defense—by resembling species that are dangerous or poisonous—or, more rarely, as a form of camouflage for launching an attack. Mimicry is not only visual: some insects mimic the behavior and even smell of others, so that they can enter their colonies and feed on their young without being attacked.



UNLIKELY MEAL

A postman butterfly caterpillar's first line of defense is its resemblance to bird droppings. If another animal approaches, it waves its menacing-looking “horns,” which give off an unpleasant smell.

mate. Unlike vertebrates, invertebrate hearing apparatus is not necessarily located on the head. Grasshoppers, for example, have eardrums in their abdomen, while crickets have them on their legs. The same is true of taste organs: flies and butterflies have chemical sensors on their feet, which means they can decide whether something is edible simply by standing on it.

Mobile invertebrates also need to know about their orientation and motion. Most have gravity sensors, that tell them which way up they are, and tension sensors that are triggered when they move body parts such as legs and wings.